

LM567/LM567C

Tone Decoder

General Description

The LM567 and LM567C are general purpose tone decoders designed to provide a saturated transistor switch to ground when an input signal is present within the passband. The circuit consists of an I and Q detector driven by a voltage controlled oscillator which determines the center frequency of the decoder. External components are used to independently set center frequency, bandwidth and output delay.

Features

- 20 to 1 frequency range with an external resistor
- Logic compatible output with 100 mA current sinking capability
- Bandwidth adjustable from 0 to 14%

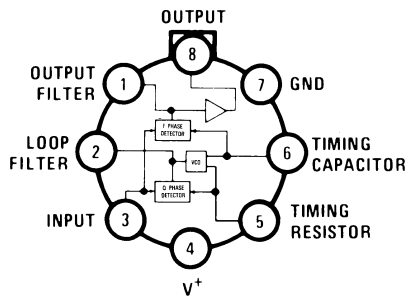
- High rejection of out of band signals and noise
- Immunity to false signals
- Highly stable center frequency
- Center frequency adjustable from 0.01 Hz to 500 kHz

Applications

- Touch tone decoding
- Precision oscillator
- Frequency monitoring and control
- Wide band FSK demodulation
- Ultrasonic controls
- Carrier current remote controls
- Communications paging decoders

Connection Diagrams

Metal Can Package

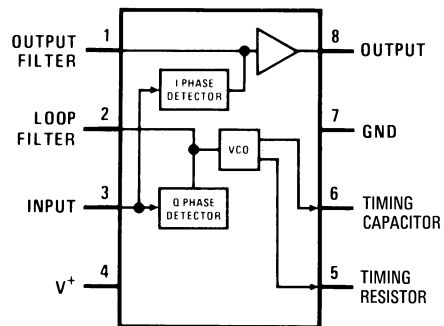


00697501

Top View

Order Number LM567H or LM567CH
See NS Package Number H08C

Dual-In-Line and Small Outline Packages



00697502

Top View

Order Number LM567CM
See NS Package Number M08A
Order Number LM567CN
See NS Package Number N08E

Absolute Maximum Ratings (Note 1)

If Military/Aerospace specified devices are required, please contact the National Semiconductor Sales Office/Distributors for availability and specifications.

Supply Voltage Pin	9V
Power Dissipation (Note 2)	1100 mW
V_8	15V
V_3	-10V
V_3	$V_4 + 0.5V$
Storage Temperature Range	-65°C to +150°C
Operating Temperature Range	

LM567H -55°C to +125°C

LM567CH, LM567CM, LM567CN 0°C to +70°C

Soldering Information

Dual-In-Line Package	
Soldering (10 sec.)	260°C
Small Outline Package	
Vapor Phase (60 sec.)	215°C
Infrared (15 sec.)	220°C

See AN-450 "Surface Mounting Methods and Their Effect on Product Reliability" for other methods of soldering surface mount devices.

Electrical Characteristics

AC Test Circuit, $T_A = 25^\circ\text{C}$, $V^+ = 5V$

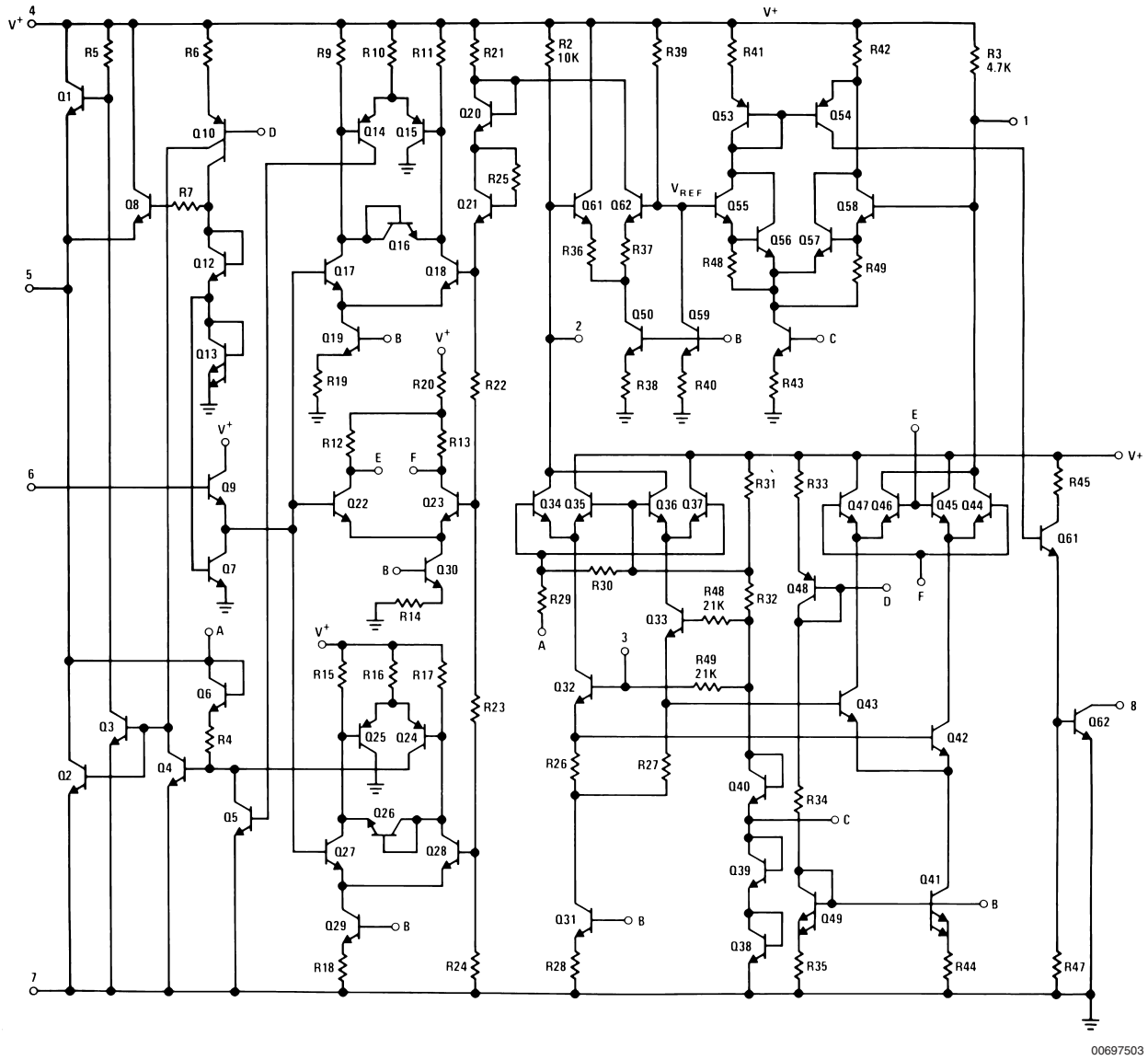
Parameters	Conditions	LM567			LM567C/LM567CM			Units
		Min	Typ	Max	Min	Typ	Max	
Power Supply Voltage Range		4.75	5.0	9.0	4.75	5.0	9.0	V
Power Supply Current Quiescent	$R_L = 20k$		6	8		7	10	mA
Power Supply Current Activated	$R_L = 20k$		11	13		12	15	mA
Input Resistance		18	20		15	20		k Ω
Smallest Detectable Input Voltage	$I_L = 100 \text{ mA}$, $f_i = f_o$		20	25		20	25	mVrms
Largest No Output Input Voltage	$I_C = 100 \text{ mA}$, $f_i = f_o$	10	15		10	15		mVrms
Largest Simultaneous Outband Signal to Inband Signal Ratio			6			6		dB
Minimum Input Signal to Wideband Noise Ratio	$B_n = 140 \text{ kHz}$		-6			-6		dB
Largest Detection Bandwidth		12	14	16	10	14	18	% of f_o
Largest Detection Bandwidth Skew			1	2		2	3	% of f_o
Largest Detection Bandwidth Variation with Temperature			± 0.1			± 0.1		%/°C
Largest Detection Bandwidth Variation with Supply Voltage	4.75–6.75V		± 1	± 2		± 1	± 5	%V
Highest Center Frequency		100	500		100	500		kHz
Center Frequency Stability (4.75–5.75V)	$0 < T_A < 70$ $-55 < T_A < +125$		35 ± 60 35 ± 140			35 ± 60 35 ± 140		ppm/°C ppm/°C
Center Frequency Shift with Supply Voltage	4.75V–6.75V 4.75V–9V		0.5 2.0	1.0 2.0		0.4 2.0	2.0 2.0	%/V %/V
Fastest ON-OFF Cycling Rate			$f_o/20$			$f_o/20$		
Output Leakage Current	$V_8 = 15V$		0.01	25		0.01	25	μA
Output Saturation Voltage	$e_i = 25 \text{ mV}$, $I_8 = 30 \text{ mA}$ $e_i = 25 \text{ mV}$, $I_8 = 100 \text{ mA}$		0.2 0.6	0.4 1.0		0.2 0.6	0.4 1.0	V
Output Fall Time			30			30		ns
Output Rise Time			150			150		ns

Note 1: Absolute Maximum Ratings indicate limits beyond which damage to the device may occur. Operating Ratings indicate conditions for which the device is functional, but do not guarantee specific performance limits. Electrical Characteristics state DC and AC electrical specifications under particular test conditions which guarantee specific performance limits. This assumes that the device is within the Operating Ratings. Specifications are not guaranteed for parameters where no limit is given, however, the typical value is a good indication of device performance.

Note 2: The maximum junction temperature of the LM567 and LM567C is 150°C. For operating at elevated temperatures, devices in the TO-5 package must be derated based on a thermal resistance of 150°C/W, junction to ambient or 45°C/W, junction to case. For the DIP the device must be derated based on a thermal resistance of 110°C/W, junction to ambient. For the Small Outline package, the device must be derated based on a thermal resistance of 160°C/W, junction to ambient.

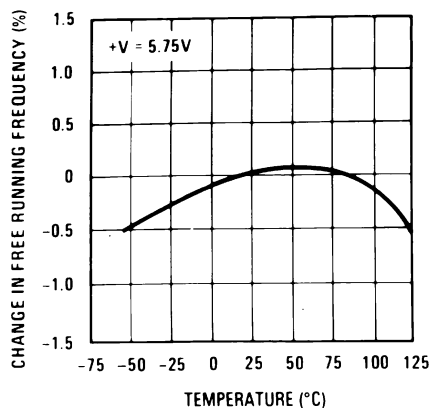
Note 3: Refer to RETS567X drawing for specifications of military LM567H version.

Schematic Diagram



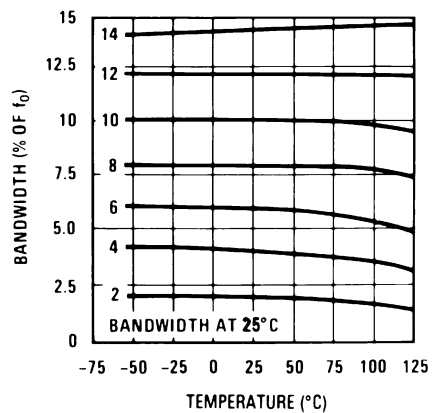
Typical Performance Characteristics

Typical Frequency Drift



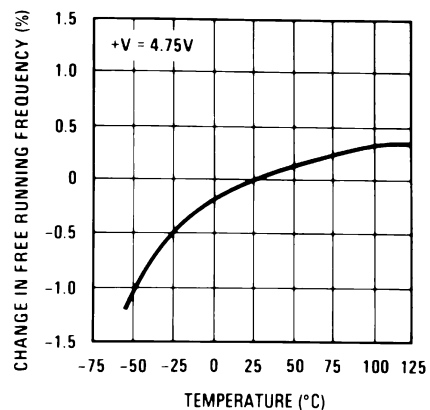
00697510

Typical Bandwidth Variation



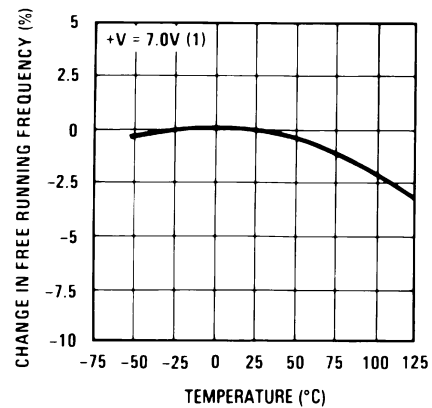
00697511

Typical Frequency Drift



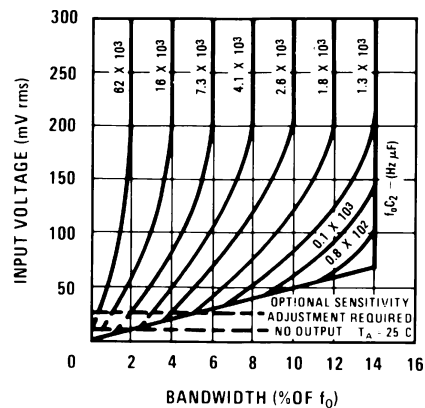
00697512

Typical Frequency Drift



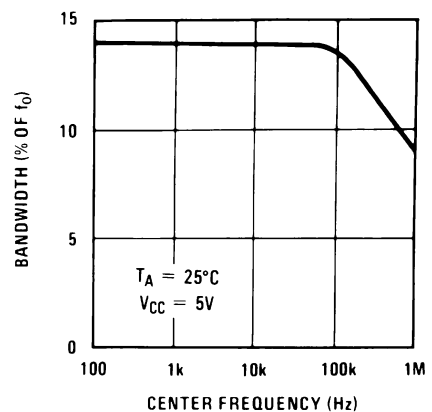
00697513

Bandwidth vs Input Signal Amplitude



00697514

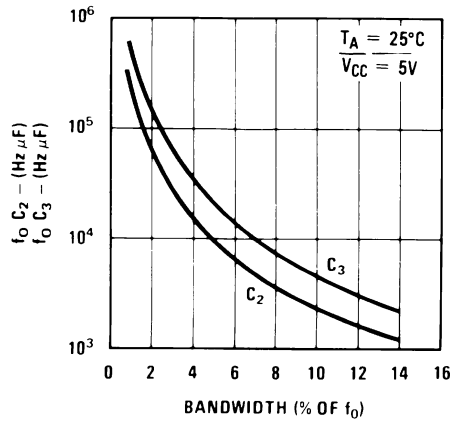
Largest Detection Bandwidth



00697515

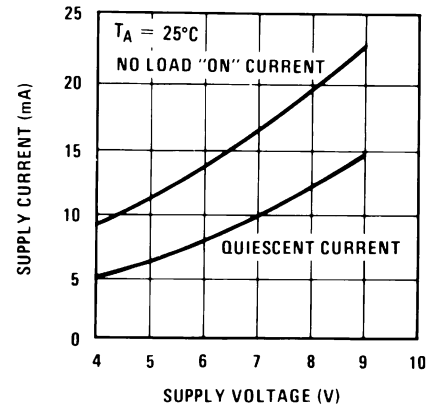
Typical Performance Characteristics (Continued)

Detection Bandwidth as a Function of C_2 and C_3



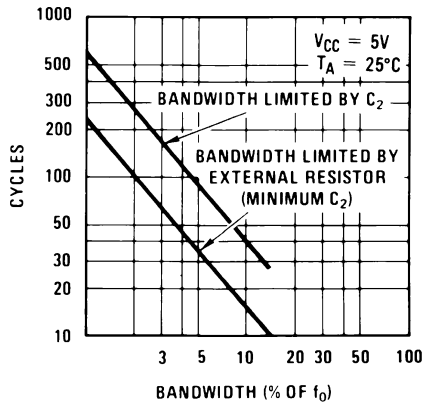
00697516

Typical Supply Current vs Supply Voltage



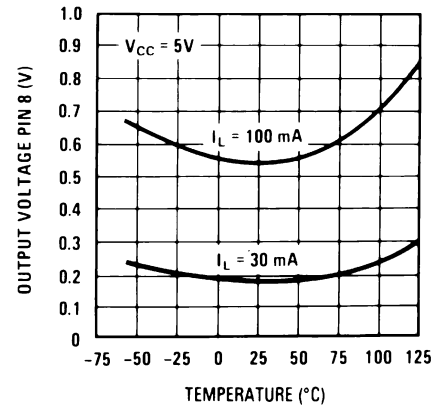
00697517

Greatest Number of Cycles Before Output



00697518

Typical Output Voltage vs Temperature



00697519

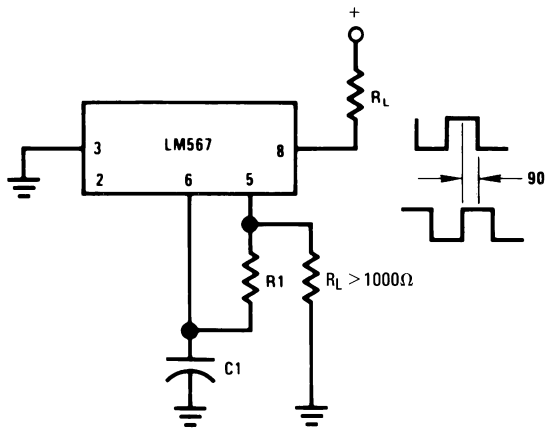
The diagram illustrates a 6-bit digital-to-analog converter (DAC) circuit. It features six LM567 precision centron oscillators, each configured with a specific frequency: 697 Hz, 770 Hz, 852 Hz, 941 Hz, 1209 Hz, and 1477 Hz. Each oscillator is biased with a 0.5 μF capacitor and a resistor network (R1, R2, R3). The outputs of these oscillators are connected to a 6-bit digital input bus, which is then processed by a DM5402 digital-to-analog converter. The DM5402 is shown as a dashed box containing six 4-input AND gates, each receiving one of the oscillator outputs and a common digital input signal. The final output of the DAC is a 6-bit digital signal, labeled DIGIT, with outputs 1 through 6.

R1	6.8 to 15k
R2	4.7k
R3	20k
C1	0.10 mfd
C2	1.0 mfd 6V
C3	2.2 mfd 6V
C4	250 mfd 6V

00697505

Typical Applications (Continued)

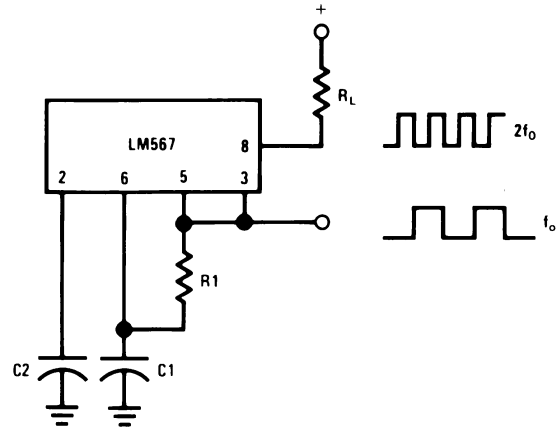
Oscillator with Quadrature Output



Connect Pin 3 to 2.8V to Invert Output

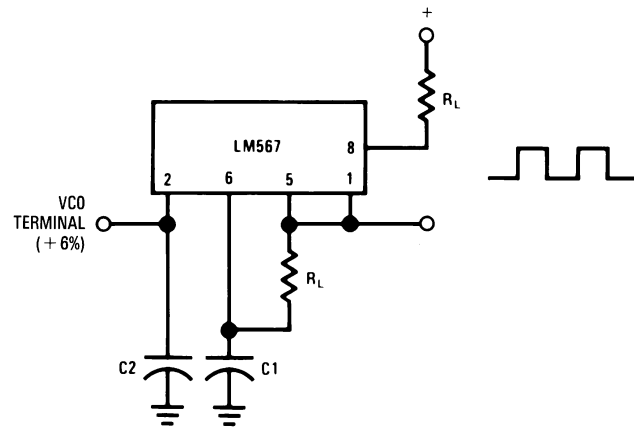
00697506

Oscillator with Double Frequency Output



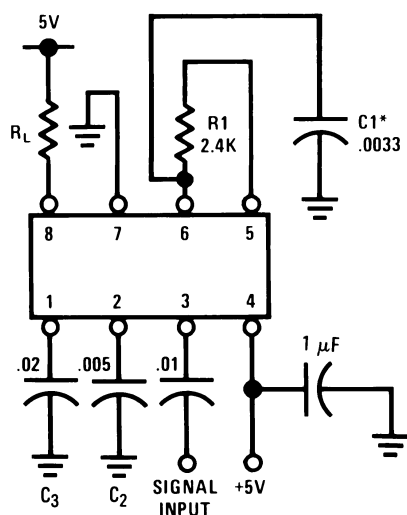
00697507

Precision Oscillator Drive 100 mA Loads



00697508

AC Test Circuit



00697509

$f_i = 100 \text{ kHz} + 5V$

***Note:** Adjust for $f_o = 100 \text{ kHz}$.

Applications Information

The center frequency of the tone decoder is equal to the free running frequency of the VCO. This is given by

$$f_o \cong \frac{1}{1.1 R_1 C_1}$$

The bandwidth of the filter may be found from the approximation

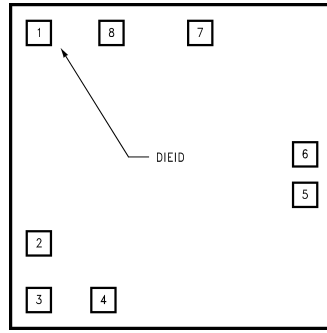
$$BW = 1070 \sqrt{\frac{V_i}{f_o C_2}} \text{ in } \% \text{ of } f_o$$

Where:

V_i = Input voltage (volts rms), $V_i \leq 200\text{mV}$

C_2 = Capacitance at Pin 2 (μF)

LM567C MDC MWC TONE DECODER



00697523

Die Layout (C - Step)

DIE/WAFER CHARACTERISTICS

Fabrication Attributes		General Die Information	
Physical Die Identification	LM567C	Bond Pad Opening Size (min)	91 μ m x 91 μ m
Die Step	C	Bond Pad Metalization	0.5% COPPER_BAL. ALUMINUM
Physical Attributes		Passivation	VOM NITRIDE
Wafer Diameter	150mm	Back Side Metal	BARE BACK
Dise Size (Drawn)	1600 μ m x 1626 μ m 63.0mils x 64.0mils	Back Side Connection	Floating
Thickness	406 μ m Nominal		
Min Pitch	198 μ m Nominal		

Special Assembly Requirements:

Note: Actual die size is rounded to the nearest micron.

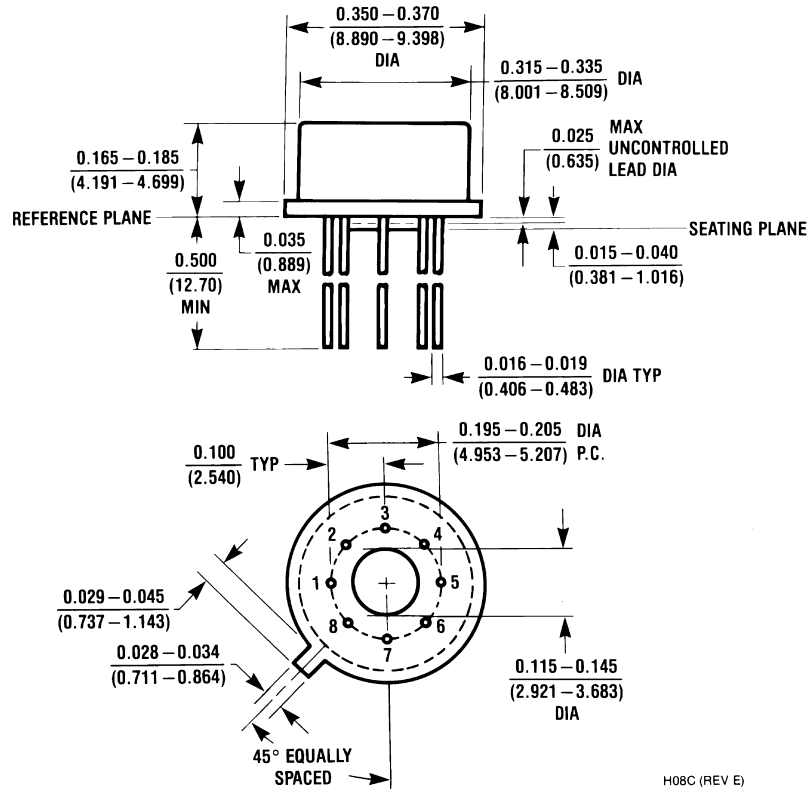
Die Bond Pad Coordinate Locations (C - Step)						
(Referenced to die center, coordinates in μ m) NC = No Connection, N.U. = Not Used						
SIGNAL NAME	PAD# NUMBER	X/Y COORDINATES		PAD SIZE		
		X	Y	X		Y
OUTPUT FILTER	1	-673	686	91	x	91
LOOP FILTER	2	-673	-419	91	x	91
INPUT	3	-673	-686	91	x	91
V+	4	-356	-686	91	x	91
TIMING RES	5	673	-122	91	x	91
TIMING CAP	6	673	76	91	x	91
GND	7	178	686	117	x	91
OUTPUT	8	-318	679	117	x	104

LM567C MDC MWC
TONE DECODER (Continued)

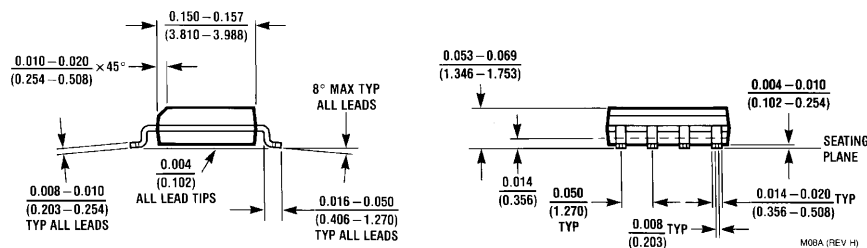
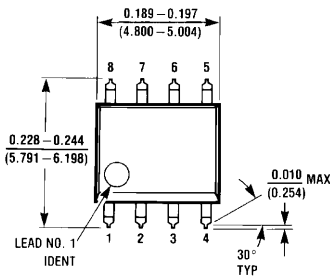
IN U.S.A	
Tel #:	1 877 Dial Die 1 877 342 5343
Fax:	1 207 541 6140
IN EUROPE	
Tel:	49 (0) 8141 351492 / 1495
Fax:	49 (0) 8141 351470
IN ASIA PACIFIC	
Tel:	(852) 27371701
IN JAPAN	
Tel:	81 043 299 2308

Physical Dimensions inches (millimeters)

unless otherwise noted

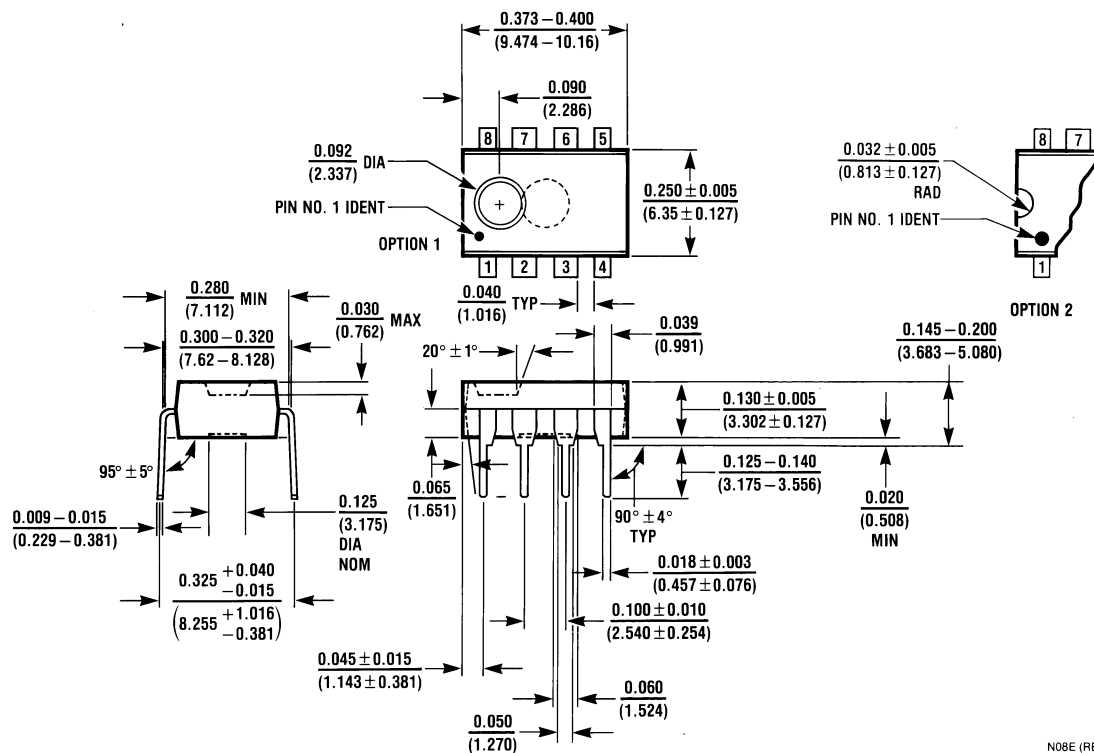


Metal Can Package (H)
Order Number LM567H or LM567CH
NS Package Number H08C



Small Outline Package (M)
Order Number LM567CM
NS Package Number M08A

Physical Dimensions inches (millimeters) unless otherwise noted (Continued)



Molded Dual-In-Line Package (N)
Order Number LM567CN
NS Package Number N08E

N08E (REV F)